

Post-War Building Assessment

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It is well-known that wars are among the most disastrous events causing significant material and human losses. Gaza has experienced numerous wars over recent years, the most intense being the war of October 7, 2023. With Gaza suffering widespread destruction, the process of reconstruction has become one of the most critical steps following the war, requiring effective strategies to assess buildings and infrastructure and to document the damages. This process faces numerous challenges, which we aim to address or, at the very least, mitigate.

We have to build a fuzzy expert system aimed at addressing these challenges by determining the damage percentage in buildings based on the input data, handling uncertainty, and providing recommendations and solutions for the existing problems.

Post-War Building Assessment Challenges:

1. Structural Safety Challenges

- **Main Factors:**

Columns and Foundations: Difficulty in accessing primary and secondary columns and foundations to assess damage, especially if covered with debris.
Invisible Damage: Potential for hidden damage within structural elements, such as deep cracks, steel corrosion, or rust.
Ceilings: Presence of cracks or tilting in ceilings, which may lead to sudden collapse.

- **Secondary Factors:**

Walls and Floors: The spread of cracks could cause sudden collapse during inspection.
Water and Sewage Drainage: Leaks cause environmental and structural damage, complicating the assessment. Evaluating the impact of leaks on structures may require specialized equipment or risk additional damage.
Electrical Network: Risk of electric shock or fire due to damaged wiring.

- **Supplementary Factors:**

Doors, Windows, and Fences: The presence of materials like wood, glass, and aluminum can obstruct the assessment process and increase the risk of injuries, such as cuts, during work.

2. Economic Constraints

- **Limited Resources:** Insufficient resources and difficulty accessing necessary machinery and equipment.
- **Repair Costs:** Repair costs might exceed the cost of demolition and reconstruction.
- **Time Factor:** Repair time could be extensive, depending on available resources and the extent of the damage. This is further complicated by the lack of alternative housing for affected residents during the repair period.

3. Assessment Risks

- **Unexploded Ordnance or Hazardous Materials:** Risk of encountering unexploded bombs or dangerous substances.
- **Falling Debris:** Danger of suspended debris collapsing during work.
- **Sudden Building Collapse:** The potential for sudden building collapse during inspection or repair.

Basic Criteria for Post-War Damage Assessment:

1. Structural Safety:

- **Main Factors:** Primary columns, secondary columns, foundations, ceilings.
- **Secondary Factors:** Walls, floors, stairs, water and sewage drainage, electrical network.
- **Supplementary Factors:** Doors, windows, fences.

2. Type of Damage:

- Major cracks, minor cracks, partial collapse, partial subsidence, noticeable tilting, slight tilting, major leaks, minor leaks, severe corrosion, slight corrosion, heavy rust, light rust, no damage.

3. Economic Analysis:

- Repair costs versus demolition, repair time.

4. Risks:

- Unexploded explosives, inactive explosives, large suspended debris, small suspended debris, active fires, remnants of stable fires, imminent collapse, uncertain collapse, no risk factors.

Reasons for Uncertainty:

1. Inaccurate Data:

- Lack of credibility and precision due to difficulty accessing buildings for inspection, either because of insufficient equipment and funding or the risks involved in entering damaged buildings, such as potential unexploded ordnance or the possibility of sudden collapse.

2. Incomplete Data:

- Lack of essential engineering data and building records, such as maps, blueprints, and information about the building's construction history.
- Loss of documents and official records, either because these buildings were unlicensed or due to their destruction by bombardment.
- Inability to reach the affected homeowners (e.g., displaced residents, those who migrated abroad, or cases where all occupants of a house were killed).

3. Conflicting Data:

- Unreliable or incorrect self-assessments due to a shortage of engineers and assessment workers (many of whom were killed or are unsafe).
- Remaining workers might lack the necessary expertise or credibility.

Methodology :

1. Fuzzy Logic

Specify the problem and define linguistic variables:

Define inputs and outputs as continuous ranges (e.g. repair cost, repair time, structural factor, partial damage level, total damage, risk level, recommendation) or discontinuous variables (e.g. structural safety, damage type, risk factor)

Define fuzzy sets for each variable: Divide each linguistic variable into fuzzy sets, such as (very high, high, medium, low, very low), allowing values to be represented with degrees of membership between 0 and 1.

Determine and visualize fuzzy sets.

2. Elicit and construct fuzzy rules

Construct IF-THEN rules based on the relationship between inputs and outputs.

Handle uncertainty and integrate uncertainty management into rules.

IF-THEN Rules:

1. Structural Safety Factor:

1. IF (Structural Safety is Primary Column)
OR (Structural Safety is Secondary Column)
OR (Structural Safety is Foundation)
OR (Structural Safety is Ceiling)
THEN (Structural Factor is Main)
2. IF (Structural Safety is Wall)
OR (Structural Safety is Floor)
OR (Structural Safety is Staircase)
OR (Structural Safety is Water Pipe)
OR (Structural Safety is Sewage Pipe)
OR (Structural Safety is Electrical Network)
THEN (Structural Factor is Secondary)
3. IF (Structural Safety is Door)
OR (Structural Safety is Window)
OR (Structural Safety is Fence)
THEN (Structural Factor is Supplementary)

2. Damage Type:

1. IF (Damage Type is Major Crack)
OR (Damage Type is Partial Collapse)
OR (Damage Type is Significant Tilt)
OR (Damage Type is Major Leak)
OR (Damage Type is Severe Corrosion)
OR (Damage Type is Heavy Rust)
THEN (Partial Damage Level is Deep)
2. IF (Damage Type is Minor Crack)
OR (Damage Type is Minor Leak)
OR (Damage Type is Mild Corrosion)
OR (Damage Type is Light Rust)
OR (Damage Type is Partial Settlement)
OR (Damage Type is Mild Tilt)
THEN (Partial Damage Level is Surface)
3. IF (Damage Type is No Damage)
THEN (Partial Damage Level is Nothing)

3. Total Damage Based on Structural Factor and Damage Level:

1. IF (Structural Factor is Main)
AND (Partial Damage Level is Deep)
THEN (Total Damage is Severe)
2. IF (Structural Factor is Main)
AND (Partial Damage Level is Unknown)
THEN (Total Damage is Severe)
3. IF (Structural Factor is Main)
AND (Partial Damage Level is Surface)
THEN (Total Damage is Moderate)
4. IF (Structural Factor is Secondary)
AND (Partial Damage Level is Deep)
THEN (Total Damage is Moderate)
5. IF (Structural Factor is Secondary)
AND (Partial Damage Level is Surface)
THEN (Total Damage is Minor)
6. IF (Structural Factor is Secondary)
AND (Partial Damage Level is Unknown)
THEN (Total Damage is Minor)
7. IF (Structural Factor is Supplementary)
AND (Partial Damage Level is Deep)
THEN (Total Damage is Minor)
8. IF (Structural Factor is Supplementary)
AND (Partial Damage Level is Surface)
THEN (Total Damage is Safe)
9. IF (Structural Factor is Supplementary)
AND (Partial Damage Level is Nothing)
THEN (Total Damage is Safe)
10. IF (Structural Factor is Supplementary)
AND (Partial Damage Level is Unknown)
THEN (Total Damage is Safe)

4. Total Damage Based on Repair Cost and Time:

1. IF (Repair Cost is Very High)
AND (Repair Time is Very Long)
THEN (Total Damage is Severe)
2. IF (Repair Cost is Very High)
AND (Repair Time is Unknown)
THEN (Total Damage is Severe)
3. IF (Repair Cost is High)
AND (Repair Time is Long)
THEN (Total Damage is Moderate)
4. IF (Repair Cost is Medium)
AND (Repair Time is Long)
THEN (Total Damage is Moderate)
5. IF (Repair Cost is Low)
AND (Repair Time is Medium)
THEN (Total Damage is Minor)
6. IF (Repair Cost is Low)
AND (Repair Time is Short)
AND (Partial Damage Level is Surface)
THEN (Total Damage is Safe)
7. IF (Repair Cost is Very High)
AND (Repair Time is Medium)
AND (Partial Damage Level is Deep)
THEN (Total Damage is Severe)
8. IF (Repair Cost is Medium)
AND (Repair Time is Very Long)
AND (Partial Damage Level is Surface)
THEN (Total Damage is Moderate)
9. IF (Repair Cost is High)
AND (Repair Time is Medium)
AND (Partial Damage Level is Deep)
THEN (Total Damage is Severe)

5. Risk Level Based on Risk Factor:

1. IF (Risk Factor is Unexploded Ordnance)
OR (Risk Factor is Large Hanging Debris)
OR (Risk Factor is Active Fire)
OR (Risk Factor is Imminent Collapse)
THEN (Risk Level is High)
 2. IF (Risk Factor is Inactive Ordnance)
OR (Risk Factor is Small Hanging Debris)
OR (Risk Factor is Stable Fire Residues)
OR (Risk Factor is Uncertain Collapse)
THEN (Risk Level is Medium)
 3. IF (Risk Factor is No Risk Factors)
THEN (Risk Level is Safe)
- #### 6. Risk Level Based on Structural Factor and Partial Damage Level:
1. IF (Structural Factor is Main)
AND (Partial Damage Level is Nothing)
THEN (Risk Level is High)
 2. IF (Structural Factor is Secondary)
AND (Partial Damage Level is Unknown)
THEN (Risk Level is Low)
 3. IF (Structural Factor is Secondary)
AND (Partial Damage Level is Nothing)
THEN (Risk Level is Safe)
 4. IF (Structural Factor is Supplementary)
AND (Partial Damage Level is Nothing)
THEN (Risk Level is Safe)
 5. IF (Structural Factor is Supplementary)
AND (Partial Damage Level is Unknown)
THEN (Risk Level is Safe)

7. Recommendations Based on Total Damage and Risk Level:

1. IF (Total Damage is Severe)
AND (Risk Level is High)
THEN (Recommendation is "Immediate evacuation and no entry, followed by demolition and reconstruction")
2. IF (Total Damage is Severe)
AND (Risk Level is Medium)
THEN (Recommendation is "Temporary evacuation and securing the site, followed by demolition and reconstruction")
3. IF (Total Damage is Moderate)
AND (Risk Level is High)
THEN (Recommendation is "Evacuate affected areas, monitor using drones, and perform urgent repairs")
4. IF (Total Damage is Unknown)
AND (Risk Level is High)
THEN (Recommendation is "Immediate evacuation and the use of drones for assessment")
5. IF (Total Damage is Moderate)
AND (Risk Level is Medium)
THEN (Recommendation is "Secure and monitor the site using drones, and perform repairs as soon as possible")
6. IF (Total Damage is Minor)
AND (Risk Level is Safe)
THEN (Recommendation is "Perform routine maintenance and minor improvements")
7. IF (Total Damage is Safe)
AND (Risk Level is Safe)
THEN (Recommendation is "Regular use without restrictions")
8. IF (Total Damage is Moderate)
AND (Risk Level is Safe)
THEN (Recommendation is "Comprehensive repair before use")
9. IF (Total Damage is Moderate)
AND (Risk Level is Unknown)
THEN (Recommendation is "Minimize the use of the building to the bare minimum and conduct an additional assessment of the risk level")
10. IF (Total Damage is Minor)
AND (Risk Level is High)
THEN (Recommendation is "Perform minor repairs to prevent increased risk")
11. IF (Total Damage is Minor)
AND (Risk Level is Unknown)
THEN (Recommendation is "Continue usage with regular site monitoring and risk assessment")

3. Control System

Integrate fuzzy rules into a control system to assess various inputs (e.g., structural safety, risk factor or damage type) and infer outputs (e.g., risk level, total damage or recommendation).

4. Evaluate and tune the fuzzy expert system

The system performs a fuzzy inference process:

1. Fuzzification: Converts numerical inputs into fuzzy values.
2. Inference: Applies fuzzy rules to derive conclusions.
3. Defuzzification: Converts fuzzy outputs into clear values or direct recommendations.

Results:

After initializing the system, the system ask the user to choose if he want to enter a numeric value or a linguistic value, then he should enter the inputs (structural safety, damage type, repair cost, repair time, risk factor), if one of these inputs is missing, he will enter "Unknown". If he enter invalid values, the four test cases will be implemented.

Suppose that the user enters an invalid value, the results will be like that:

Test the system using 4 different cases...

Test case 1: {'structural_safety': 'Door', 'damage_type': 'No Damage', 'repair_cost': 'Very Low', 'repair_time': 'Very Short', 'risk_factor': 'No Risk Factors'}

Simulation results...

Total Damage Level (Ratio): 0.10

Recommendation (Ratio): 0.10

Total Damage Level: Safe

Recommendation: Regular use without restrictions

When the building has no damage doors, the repair cost is very low, the repair time is very short, and there is no risk factors: the logical result is that the building is safe and there is no need to any reclamation, so the total damage level is 10% (safe) and the recommendation is: Regular use without restrictions.

Test case 2: {'structural_safety': 'Wall', 'damage_type': 'Mild Corrosion', 'repair_cost': 'Low', 'repair_time': 'Short', 'risk_factor': 'No Risk Factors'}

Simulation results...

Total Damage Level (Ratio): 0.24

Recommendation (Ratio): 0.20

Total Damage Level: Minor

Recommendation: Minimize the use of the building to the bare minimum and conduct an additional assessment of the risk level

When the building has mild corrosion on walls, the repair cost is low, the repair time is short, and there is no risk factors: the logical result is that the building has minor damage and there is a need to provide minor performs, so the total damage level is 24% (minor) and the recommendation is: Minimize the use of the building to the bare minimum and conduct an additional assessment of the risk level.

Test case 3: {'structural_safety': 'Staircase', 'damage_type': 'Minor Crack', 'repair_cost': 'Medium', 'repair_time': 'Long', 'risk_factor': 'Inactive Ordnance'}
Simulation results...

Total Damage Level (Ratio): 0.50

Recommendation (Ratio): 0.60

Total Damage Level: Moderate

Recommendation: Immediate evacuation and the use of drones for assessment

When the building has minor crack on staircase, the repair cost is medium, the repair time is long, and there are inactive ordnances in the place: the logical result is that the building has moderate damage and there is a need to promote and rehabilitate the building, so the total damage level is 50% (moderate) and the recommendation is: Immediate evacuation and the use of drones for assessment.

Test case 4: {'structural_safety': 'Primary Column', 'damage_type': 'Major Leak', 'repair_cost': 'Unknown', 'repair_time': 'Very Long', 'risk_factor': 'Unexploded Ordnance'}

Simulation results...

Total Damage Level (Ratio): 0.85

Recommendation (Ratio): 0.97

Total Damage Level: Severe

Recommendation: Immediate evacuation and no entry, followed by demolition and reconstruction

When the building has partial collapse on foundation, the repair cost is missing, the repair time is very long, and there are unexploded ordnance in the place: the logical result is that the building has severe damage and there is a need to remove and rebuild the building, so the total damage level is 85% (severe) and the recommendation is: Immediate evacuation and no entry, followed by demolition and reconstruction.

Comments:

Building an fuzzy expert system considers challenges of post war building assessments has many challenges:

Challenge:

Setting appropriate labels (linguistic variables) in the inputs and ensuring they are correctly mapped to the outputs. This is crucial for maintaining consistency and accuracy in the system's assessment.

Solution:

Develop a function that assigns meaningful labels (e.g., "Low," "Medium," "High") to each input variable based on predefined thresholds or membership functions.

Use a structured mapping process (e.g., a dictionary or rule matrix) to correlate input labels to their corresponding outputs.

Challenge:

Managing uncertainty:

In some cases, data may not be fully available (e.g. unknown values for variables).

Not handling uncertainty properly can lead to inaccurate results.

Solution:

Dealing with "unknown values" within rules using default conditions.

Use fewer, more general rule sets to cover multiple scenarios.

Additional improvements:

- Integrate machine learning techniques to improve rule generation based on historical data.
- Use tools like TensorFlow Fuzzy to speed up computations in fuzzy systems.

Conclusion:

This project aims to handle uncertainty for missing or imprecise data that often associated with damage assessments post wars. By using fuzzy logic, the system can provide recommendations based on practical knowledge and expertise in a flexible and efficient manner.

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